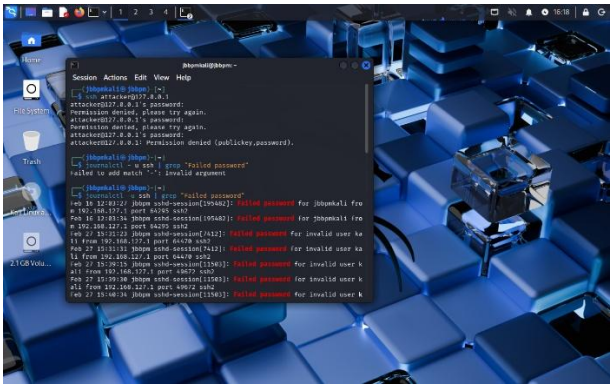
Linux Server Hardening Project

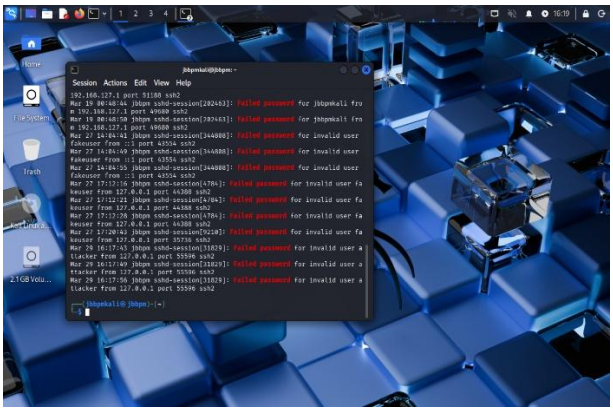
Tools Used: KaliLinux Virtual Machine

Detection

A potential brute force attack was detected targeting SSH authentication services. Multiple failed login attempts were observed from a single source, followed by a successful login event, indicating a possible compromise.

Log analysis using *journalctl,* revealed multiple failed login attempts:





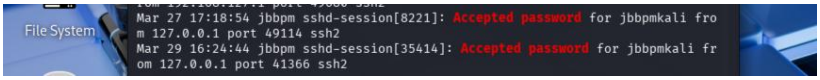
Observations:

- Multiple Failed attempts
- The last 3 Failed attempts came from the same port – 127.0.0.1

Analysis

Further investigation showed:

- Repeated attempts against invalid user accounts
- A successful login event following failed attempts



Containment

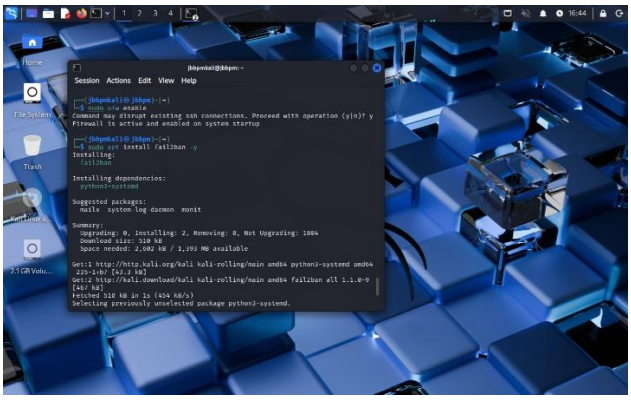
- Blocked attacker IP using firewall rules using command – iptables -A INPUT -s 127.0.0.1 -j DROP
- Terminated active sessions

Eradication

- Verified no unauthorized users in system
- Checked SSH key access
- Confirmed no persistence mechanisms

Recovery

- Reset system credentials
- Enabled firewall protection using sudo ufw
- Installed intrusion prevention tool (fail2ban)



Lessons Learned

- Lack of login attempt restrictions increases attack risk
- Monitoring logs is critical for early detection
- Intrusion prevention tools should be implemented